FAICT - The Fact AI Checking Tool

CS410 – Feasibility Presentation

Presented by:

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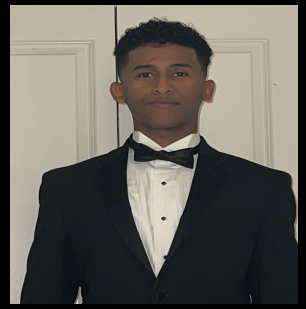

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Abstract

- Misinformation spreads rapidly across social media, news platforms, and online communities (especially with the rise of AI-generated content)
- Current fact-checking tools are often slow, limited to specific topics, and/or primarily focused on text verification.
  - *AI/chatbot solutions fail to provide a reliable, all-in-one system*
- Solution: Faíct verifies information in real-time across text, audio, and video by extracting claims and analyzing them against reliable sources
- Provides credibility assessments along with cited sources & AI-generated content detection.
- Helps users quickly identify misinformation and make informed decisions.

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The Misinformation Problem

The Core Issue

Internet users are subjected to rising levels of misinformation from online sources and platforms.

The AI Factor

The proliferation of AI-generated posts, articles, and social media accounts only worsens the problem.

Verification Gaps

Content verification features offered by websites and platforms are limited in scope and accessibility.

The Bottom Line

Users lack a universal and on-demand method for quickly verifying information they encounter on the internet.

Anatomy of the Misinformation Problem

The Scale

The spread of misinformation is rampant in everyday web browsing and social media usage.

Limits of Current Solutions

- Platform Exclusivity (limited to specific platforms)
- Popularity Bias (verification only occurs on highly viewed posts)

The User Burden

Manual validation is time consuming and confusing, requiring online searches and cross-referencing sources.

Content Generation

Manual fact checking is outpaced by the scale at which fake content is being generated from AI tools..

Who's Most Affected?

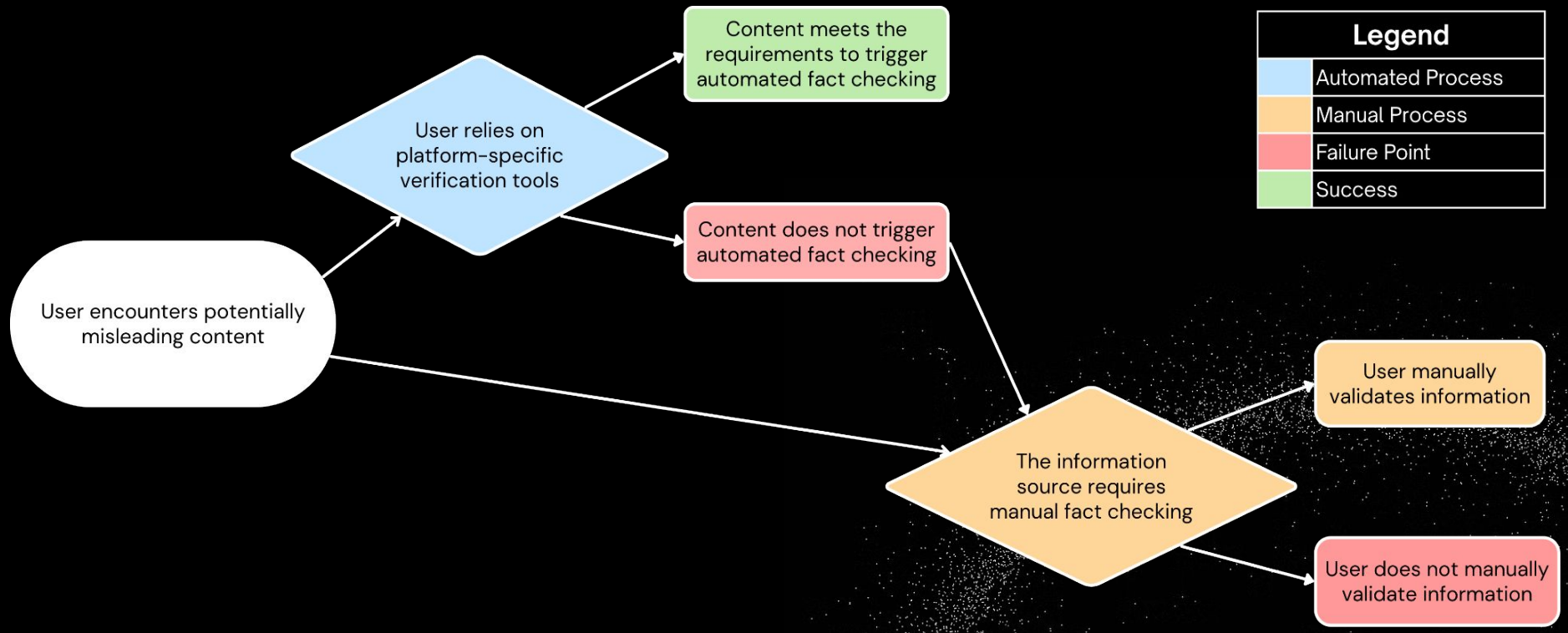
- Industry professionals that rely on real-time content verification:
  - Journalists
  - Content creators
  - Moderators
- Organizations with strict validity requirements:
  - Academic institutions
  - Research facilities
  - Risk Management Teams

Solution Statement

- Cross-platform fact checking across text, links, and social media posts, allowing users to verify information
- Real-time speech-to-text fact checking, enabling live verification of podcasts, videos, and conversations as they happen
- AI-generated content detection across text, images, and video
- Credibility scoring, providing clear ratings supported by trusted sources and clear explanations
- Fast verification, designed to assist and speed up the human fact-checking process

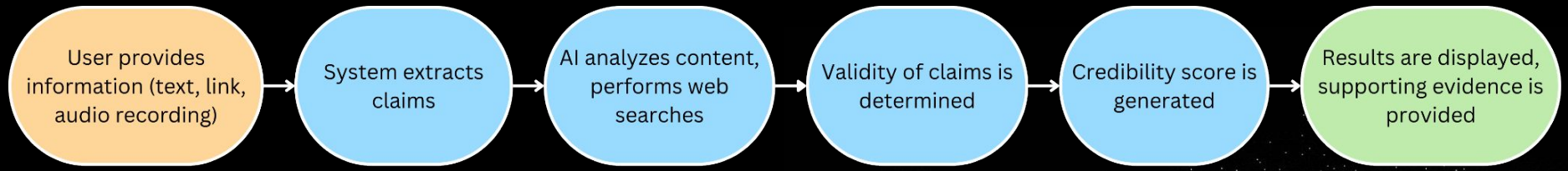
Current Process Flow

The current process is often slow, confusing, and time consuming.



Solution Process Flow

Our proposed solution is automated, faster, and provides a clear, evidence-based credibility assessment



Legend	
	Automated Process
	Manual Process
	Failure Point
	Success

What the Software WILL do:

- Accept text, links, and social media content
- Process live speech (speech → text → verification)
- Convert video → transcript → fact check
- Analyze claims using AI + verified trusted sources
- Provide credibility scores (true, misleading, false, uncertain)
- Display supporting evidence
- Detect AI-generated content

What the Software WILL NOT do:

- Guarantee 100% accuracy
- Replace human verification
- Automatically remove content
- Store sensitive user data long-term
- Provide political opinions

Competition Matrix

Feature	FAICT	Community notes	Snopes & Politics/News	Google Fact Check Tools	ChatGPT/LLMS	AI Detection Services
Text-Based Fact Checking	✓	✓	✓	✓	✓	✗
Real-time Audio Fact Checking	✓	✗	✗	✗	✓	✗
Video Upload Fact Checking	✓	✓	✗	✗	✗	✗
AI Content Detector	✓	✓	✗	✗	✓	✓
Cross Platform Use	✓	✗	✗	✓	✓	✓
Allows User Input?	✓	✓	✗	✓	✓	✓
Platform Speed	Fast	Slow	Slow	Medium	Fast	Fast

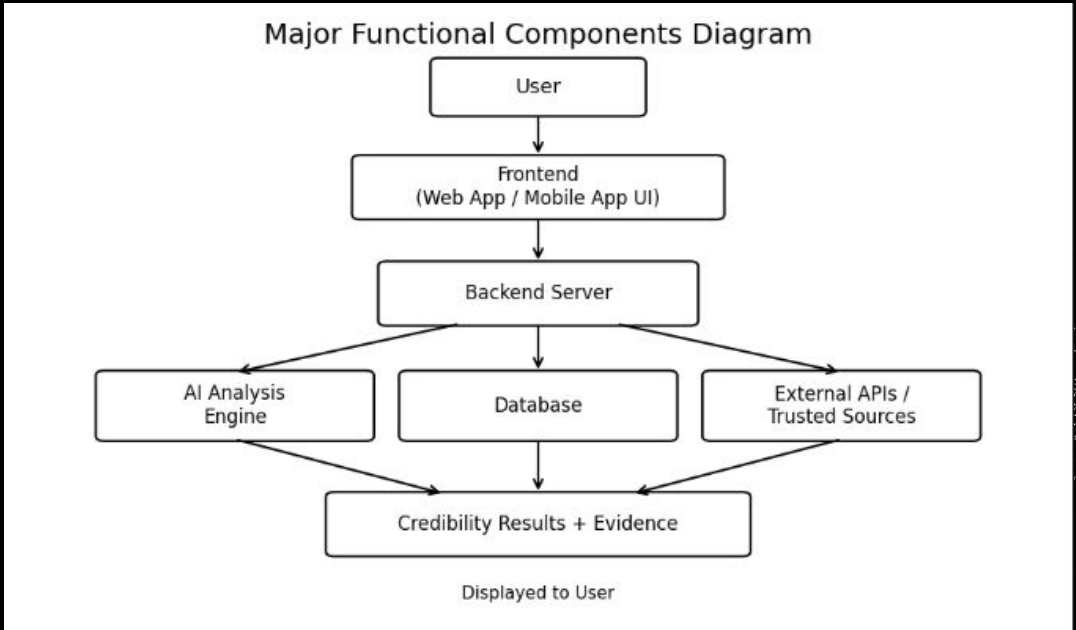
Development Tools

- IDE: VS Code
- Version Control: Git & GitHub
- Integration/Deployment: GitHub Actions
- Backend: Python
- Frontend: HTML, CSS, JavaScript
- Database: PostgreSQL / MongoDB
- Testing: PyTest

Major Functional Components

- Frontend Interface (User Input & Results Display)
- Backend Server
- AI Model (Claim analysis and classification)
- Speech-to-text module
- Video-to-transcript processing module
- AI detection module
- Database (Stores user data and results)
- External APIs (search + fact sources)

Functional Components Diagram



Risks

Risk Matrix

Risk	Type	Why It Matters	Mitigation
Data Privacy & User Input Exposure	Security	Exposes user data, reduces trust	Encrypt data, minimize storage, anonymize inputs
Model Manipulation	Technical	Produces incorrect or misleading results	Input validation, filtering, adversarial testing
Incorrect AI Assessments	Technical	Can spread misinformation	Use multiple sources, confidence scores, human review
Bias in Training Data	Technical	Skews credibility results	Use diverse datasets & credible sources with continuous evaluation
Over-Reliance on AI	Customer	Users may blindly trust outputs	Show explanations, encourage verification
System Scalability	Technical	Slows or crashes system under load	Cloud scaling, caching, optimization
Legal/Ethical Concerns	Legal	Risk of legal or reputational damage	Disclaimers, transparency

The Risks (Security)

Data Privacy and User Input Security

Users may paste sensitive data (e.g., messages or posts) into the tool.

If not properly secured, this data could be exposed through breaches or interception. Strong encryption and secure data handling are essential to protect user privacy.

Model Manipulation

Attackers may try to manipulate the AI with crafted inputs or repeated probing.

These tactics can expose weaknesses and cause incorrect credibility assessments.

This reduces reliability and may allow misinformation to go undetected.

The Risks

False Positives or Incorrect Assessments

AI can misinterpret context, sarcasm, or complex claims.

Incorrect assessments may lead users to trust and spread misinformation.

Bias in Training Data and Sources

If the system relies on limited or biased sources, its assessments may be skewed.

This can unfairly label some information as unreliable while favoring others.

Gaming of The System

If the AI relies on quantity of sources instead of true reliability, it could be tricked into validating false claims.

The Over Reliance on Automated Fact Checking

Users may over-rely on AI instead of thinking critically.

Blind trust can reduce independent verification of information.

The Solutions

- Protect user data with encryption, secure storage, and limited retention
- Mask sensitive information before analysis
- Monitor and test the model for manipulation attempts
- Use diverse, high-quality, trusted sources
- Show confidence scores and explanations with each result
- Add human review for unclear or high-risk claims
- Encourage users to verify information instead of blindly trusting AI

Monetization Plan

Potential Monetization Strategies

1

Basic Tier: Free

- Limited text/claim fact checking
- Credibility score with basic explanation
- Supported by targeted ads

2

Pro Tier: $9.99/mo

- Unlimited text fact checking
- Expanded search parameters and deeper analysis
- Live speech-to-text fact checking
- Video-to-transcript fact checking
- API usage credits for developers/integrations

3

Premium Tier: $19.99/mo

- All Pro features included
- AI-generated content detection (text, image, video)
- Advanced analysis (more parameters)
- Higher API usage credits

Market Opportunity

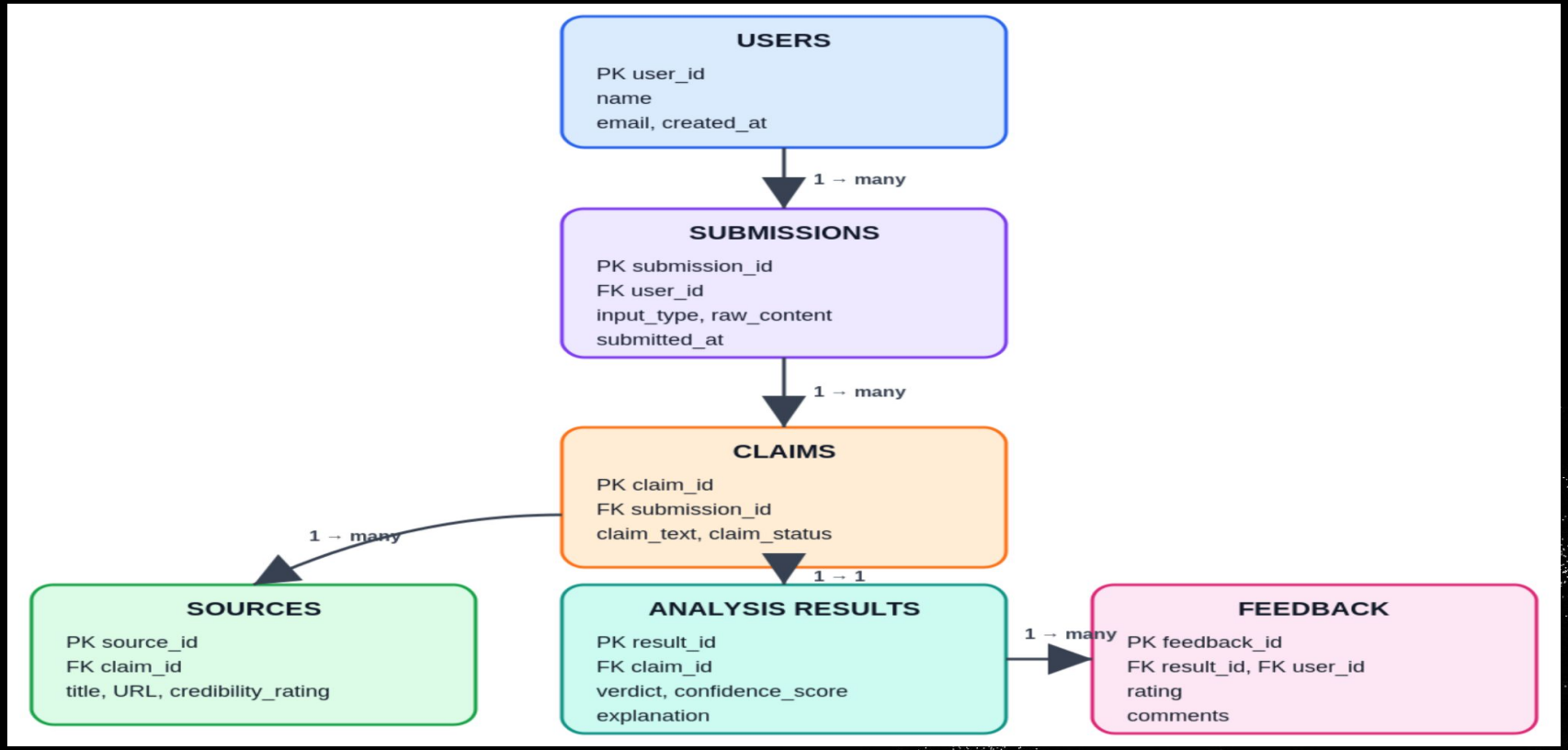
Market Size:

- ~**5.3+ billion** internet users worldwide
- ~**4.8+ billion** social media users globally
- ~**64%** of U.S. adults say misinformation causes major confusion
- ~**94%** of U.S adults check the accuracy of their news sources but not all do before sharing information
  - An overwhelming **97%** want stricter measures to limit the spread of misinformation
- Fact-checking and AI platforms show proven demand:
  - Snopes receives ~**20–22 million** monthly users
  - ~**900 million** weekly ChatGPT users

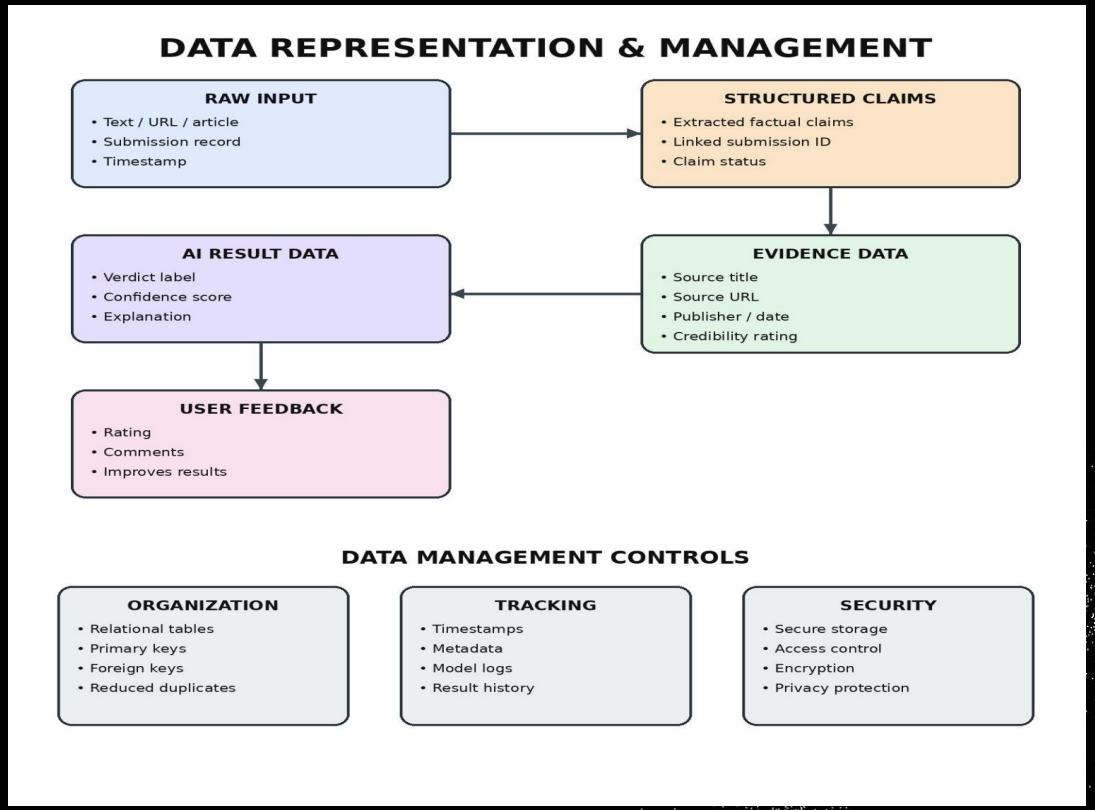
Who should be responsible for stopping the spread of misinformation?

	Overall	Democrats	Republicans
News organizations	70%	80%	54%
Individual users	62%	63%	59%
Social media platforms	59%	73%	32%
Independent watchdogs	46%	54%	31%
Government	39%	49%	18%
No one	4%	1%	8%

Database Schema



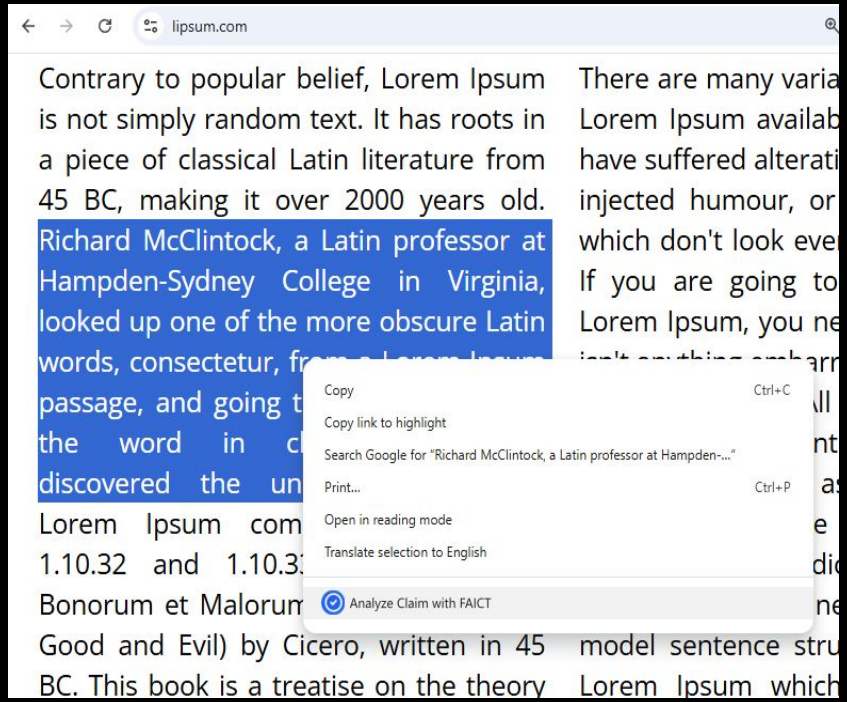
Data Representation



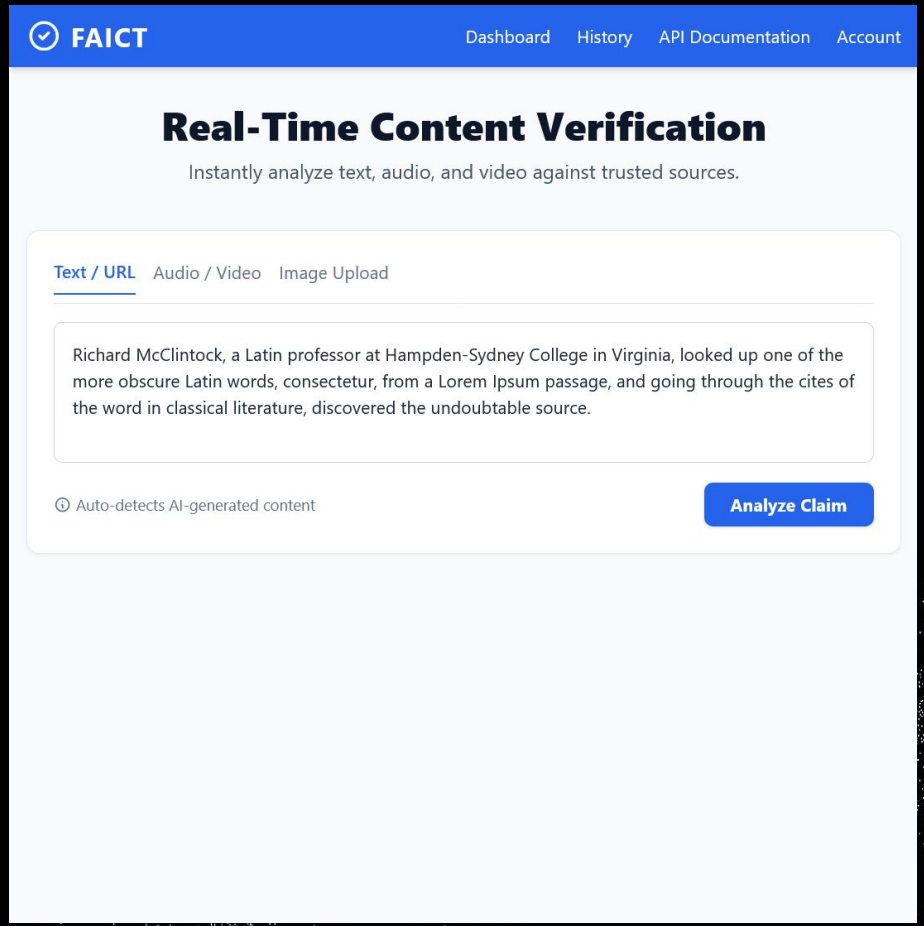
Graphical User Interface Design

- Web-based interface built with HTML/CSS/JavaScript
- Input page for text, URLs, articles, or screenshots
- Results dashboard showing verdict, confidence score, and explanation
- Evidence section showing trusted sources used in the analysis
- Feedback form for users to rate or report incorrect results
- Backend API connects the GUI to the database and AI analysis engine

Web-based GUI Mockup



The program can be accessed directly from a web page via the browser extension.



Conclusion

- Misinformation will continue to be a growing issue due to the bloom of AI and social media.
- Current tools are often platform specific, limited to viral content, and require time-consuming manual verification
- Our solution provides fast, real-time, cross-platform verification, including live speech-to-text fact checking, video analysis, and AI-generated content detection
- Delivers evidence-backed credibility insights using trusted sources
- Helps users make informed decisions and reduce misinformation